

LINEAR RANKING TOURNAMENTS

By
Michael P. Fleming

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Preface...

"Welcome to our clinic session. We are going to discuss and then experience a unique tournament that I believe is indispensable for you as a Coach and Physical Educator. First let me begin with a story:

Last year we had twenty-four players trying out for the Canadian Junior National Team and we'd been in three - sessions a day training camp for three days, so they were really sore and tired. Well, they went to bed expecting a full nights' sleep, but we gave them a surprise wake-up at 3:00 in the morning, and hustled 'em down to the gym to play in a special kind of tournament created for the C.V.A. by a British Columbia mathematician named James Bjerring that would rank order them from number 1 to 24 based on their playing ability. We told everyone their ranking would be a critical part of determining which twelve players would be selected or cut. We intentionally set up the situation to evaluate a lot of things beyond just skill. We needed to know how they could play under stress, without sleep - like simulating jet lag - hurt'n and tired, because of the fierce international schedule we faced and because you can't fake your real character at 3:30 A.M.!

Now, when choosing a team it's easy to know the top players (say one thru ten) and the poorest (like, fifteen thru twenty-four), but the critical numbers 'on the bubble' from eleven to fourteen are the toughest to figure. Based on a large variety of criteria, we had our coaching staff (that had all four 'Master Coaches' in Canada) list all the candidates top to bottom and then we discussed the order at length and came up with a single 'ranking' list. After playing the 'Bjerring Tournament' we compared the final order with the order of our list and found that they were nearly identical, except our number fourteen had "played" her way to number twelve, earning a spot on the roster. There's no substitute for pure competition for athletes and that's why I know you will value the process you will experience today called the 'Jim Bjerring Linear Ranking Tournament'".

*... Vic Lindal
Master Coach - Canadian Volleyball Association*

In the Spring of 1972 I attended a Level I Canadian Volleyball Coaches Association clinic in Castlegar, British Columbia. I was just a rookie coach and it was my first volleyball clinic. At the time there were four "Master" level coaches in Canada and three of them happened to be at that clinic. In retrospect (after all the decades in volleyball and hundreds of clinics/camps) I realize what a wonderful job they did. And what a fun time we had.

It was my first experience with a Linear Ranking Tournament. As a part of the process on Saturday evening we talked about and then played in, a brief "Bjerring Ranking Tourney". Like anything new it was confusing at first but after a short while it all came clear: the tournament pairings are based on the premise that everyone plays "with" and "against" everyone else the same number of times. So that session we were numbered off and we then played in a variety of pairings, and by the end of the schedule our entire group was rank ordered from best to poorest by playing ability! I took notes on the numbers we used that evening and figured the single set that I came away with would work just fine, if only I was lucky enough to have the exact right number of players to match. Then a couple weeks later, to my surprise, a packet arrived from one of the clinicians. Vic Lindal (at that time, Director of Player Development for B.C.) had sent me Linear Ranking Tournament numbers for 4's, 5's, 7's, 8's, 9's, 12's, 16's, and 25's... the eventual core of this work. Over the ensuing years of coaching volleyball I applied those numbers, of course to volleyball, then to my Physical Education classes for badminton, pickleball, basketball, tennis, etc. Gradually, I began to use them just to mix kids in a variety of activities, sometimes merely for social activities or for group dynamics; to be sure each person got to work together and be involved with everyone else.

I streamlined the administrative process and developed forms so players themselves could run their own tourney.

As I became involved with U.S.A. Volleyball as one of the Original Cadre in the Coaches Accreditation Program and began sharing information as a clinician, I always made it a point to give fellow coaches my "Linear Ranking stuff". It usually seemed a rushed affair that was more confusing than it ought to be. And so, I decided to put together all the Ranking Concepts and information clearly, in a single complete package. Being a thorough person, I invested a bit of time as an amateur detective and tracked down the real James Bjerring near Victoria, B.C. and asked if he would grant permission for me to publish his contribution to the volleyball world. Not only did he give permission, but he was kind enough to send a whole series of new (never before released!) tourney schedules. Exciting! Though they were the mathematically "shorthand" versions I gladly have converted them into complete, ready-to-use, "longhand" versions and have included every single one. Even the Triples for 19 with its resulting 228 pairings! Perhaps some of his creations may never be used because of the time needed to contest such lengthy pairings, but nonetheless, I have made every effort to put together the total, composite source: it's ALL here.

So, thanks to James Bjerring - and it's pronounced "Beer - ing", to Vic Lindal and to my true friend Bill Neville for sharing with me the skeleton of this concept. I share with you the whole body of work.

In Sport,

Mike Fleming

GENERAL INFORMATION

Introduction:

Mathematician James H. Bjerring has designed a special tournament that will obtain a linear ranking (poorest to best) of players within a closed group. (For us a *closed group* is any group of individuals that competes together exclusively, without addition of any other outside players.) Pairings of individuals within the tournament are made so that each person plays exactly the same number of times with and against everyone else in their group. In doubles they usually play with each player once as a partner and against every player twice as an opponent. In triples they usually play with each player twice as a partner and against every player three times as an opponent. By doing so, and keeping track of wins/loses and points-differential, rankings can be made. Mathematically it is possible to play reasonable length tournaments for the following numbers of players: 4, 5, 8, 12, 16, and 25 (for doubles), and 7 and 9 (for triples). Additionally, other numbers which require a great deal of time and multiple pairings are provided herein (they are new formulas previously unavailable).

All information from Mr. Bjerring is used with his permission.

Scoring:

Each player is associated with two numbers. The first is the individuals' number of games won. The second is the individuals' total from point differentials (points for, minus points against) in each game played. After all games are played, simply rank order the players by their win-loss records (win-loss is the most important factor) and if their records are identical use their point total to break the tie. Their overall point difference is only used for tie-breaking. If a tie still exists then the higher ranked player remains ahead of the tied lower ranked player.

Discussion:

Linear ranking tournaments can be used in a variety of situations: in athletics, physical education, and in recreational activities. They can be applied not only to volleyball (for which they were designed), but will work for any activity that can possibly be played as doubles or triples to rank order people by playing ability for skill evaluation. Such tournaments have been used to determine school, community, city, and State/Provincial individual top-ranked players. They can also serve in education, business and workplace situations to mix departments or groups of people for meetings and even social activities.

There are many implications in playing such a tourney. Perhaps the most valuable quality is the complete mix of partners and opponents. In that very process, the dominance of talented friends or players who would prefer to "pair-up" is shattered. Likewise, "cliques" that people might prefer are broken up and new combinations are experienced. As an outgrowth of random pairing, new friends and social relationships develop. Secondly, the "ranking" nature of the tournament creates meaningful competition in individual terms: even though playing with and against everyone in the group, each individual still achieves personal success within the group format. The fact that players are ranked by their performance is not lost on people. Rather than evaluation coming from a teacher/coach/supervisor, the outcome is determined by how the player plays, rather than by opinion. Additionally, the players compete in doubles or triples using the skills of the game instead of some abstract or "closely related" test... they do the very activity that is being evaluating. Of course the coach/teacher/supervisor can observe and evaluate all the individual skills and attitudes of players while the process proceeds.

When a timed format is used many important personality facets are revealed because with time dwindling away numerous scenarios occur. Does the entire team that is losing give up? Do some players give up when losing? How do their teammates react to a "quitter" in their group? What kind of interaction occurs between players in a losing situation: are they supportive, intense, hustling, flippant, etc. When trailing with time running out do all, or some, players fight hard to at least narrow the gap between their score and their opponents' score (knowing that the eventual point differential may be critical)? Similarly, on the winning side, do players "pour it on" (trying to inflate the differential)? Do they unethically slow down the game to try to run out the clock? Do they become cocky, overbearing, obnoxious, overconfident, etc. Obviously an observer can learn about much more than merely skills through the Bjerring process. It is precious information to know how well a player competes, performs, cooperates, persevere, etc., under stress.

Finally, as players become ranked through the process, they can be grouped to play with others of similar skill level. Such homogeneous groupings create opportunities to compete and learn at appropriate pace and enable players to have more fun.

Special Rules Specific to Volleyball:

This type of tournament has been used at National level volleyball training and tryout camps in many countries. The results indicate that this kind of tournament has a very high validity for ranking volleyball players. The following rules apply only to a Jim Bjerring Tournament.

1. Use a court 4.5 meters wide and 18 meters long.
 2. If there is a player "out", they officiate and supply balls.
 3. Seven minutes is the usual minimum amount of time for games and if you have time, play to 11 or 15 points.
 4. Servers and serve receivers must be rotated.
 5. Tipping is legal.
 6. Either player can attack and use any skill of the game.
 7. You can win by one point.
 8. Optional: You may choose to remove or use antennae. Original Jim Bjerring rules called for removing them.
- Note: Doubles are preferable to triples in volleyball because in doubles, if one player is weak, that player cannot be "hidden" by the partner (as is possible in triples).

ORGANIZING AND ADMINISTERING A LINEAR RANKING TOURNEY

Organizing / administering a linear ranking tourney can be very simple and after players have experienced a few rounds they will be able to run the process easily and correctly by themselves. A description of the two methods of organization follows:

A. Single-Closed-Cell Method

This method is very easy and straight forward and is used when you have the exact number of players to match the workable tourney numbers. This is the truest, most accurate linear ranking. All players remain in a single group (a closed cell) so that everyone interacts with everyone else; especially important if you are using the results as part of a team selection. The only drawback is that for doubles you need the exact numbers (4, 5, 8, 12, 16, or 25), and for triples you ought to likewise have exact numbers (7 or 9). Although for triples you can play with one less person, namely 6 or 8, which results in three versus two in most of the match-ups. Additionally, you can choose to play with one extra player (8 or 10) by having two weak players, play together as one number. The step-by-step process is as follows:

1. Determine the number of players involved. Example: 12 (See schedule for 12 players on 3 courts.)
2. Determine how many courts can be made available (remember in volleyball it's narrow court).
Example: 3
3. Assign each player a number by entering names on scoresheet. Example: one through twelve.
4. Play all the match-ups! Example: Eight minute games / eleven time slots on courts A, B, and C.
5. Add up wins and losses. Total points to break any ties that occur in win / loss.
6. Rank players based on win / loss and if necessary, points.

B. Multiple-Closed-Cell (Sort-Sift) Method

This method can be used when you have large numbers of players or when you don't have the exact numbers to match the linear ranking tournament numbers. The step-by-step process is as follows:

1. Determine the number of players involved. Example: See below, and "Sort - Sift Graphic" on next page.
2. Determine how many courts can be made available (remember in volleyball it's narrow court).
3. Use "Sub-Set Combinations To Match Group Sizes" as a "cheat-sheet" to determine courts / groups.
4. Assign each player to a group, making sure that you distribute talent as equally as you can in each group, and enter names on scoresheet.
5. Play all the match-ups!
6. Add up wins and losses. Total points to break any ties that occur in win / loss.
7. Rank players based on win / loss and if necessary, points. Players have now been SORTED and are ready to play a second round of match-ups to SIFT themselves into complete top to bottom rankings.
8. Take the appropriate number of top ranked players and place them in a NEW top group, place middle ranking players in another new group, and so on.
9. Play all the new match-ups.
10. Check win / losses / points and rank-order the groups: the top player in the top group is #1 and the last ranked player in the bottom group is the last ranked of all players involved.

An Example: The U.S.A. National Men's team had 36 players trying out. The 12 current members of the Team were given Blue jerseys. The 12 "B" Team players were given Red jerseys. The other 12 men trying out were given White jerseys. Four Blues, 4 Reds and 4 Whites were placed in groups of 12: "A", "B" and "C". The first round of complete match-ups was played. The four top ranked players from "A", from "B" and from "C" were placed in a new group: "D". (They were almost all "Blues"). The fifth thru eighth ranked players were placed in a new group: "E". (They were almost completely "Reds"). The ninth thru twelfth ranked players were placed in a new group: "F". (Yep, nearly all "Whites"). The new groupings played a second round of complete match-ups. The resulting rankings produced a linear ranking of #1 thru #12 from "D", #13 thru 24 from "E" and #25 thru #36 from "F". You could choose to play additional rounds but would shift to a "ladder" concept after the second round and would move only a selected number of players up and down the linear scale. So if our National Team groups played a third round players "D8 thru 12" would drop down a group and players "E 1 thru 4" would move up, etc.. The "SORT - SIFT GRAPHIC" that follows illustrates the concept.

You will find that with only brief exposure, players will be able to effeciently run their own tournaments. You will need only give them a scoresheet with their names on it, a pen to fill in scores and tell them how long to play each pairing. They will compete and you will be able to observe and coach while they rank order themselves.

LINEAR RANKING TOURNAMENT

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"SORT AND SIFT" GRAPHIC

Round I

START: SORT.....

Player #	New [Ranking]
A A1	Becomes [A3]
A2	Becomes [A2]
A3	Becomes [A5]
A4	Becomes [A1]
A5	Becomes [A4]

Round II

SIFT...

Player # and [Ranking]	New [Ranking]
D 1[A1]	[D3]
2[B1]	[D2]
3[C1]	[D1]
4[A2]	[D4]
5[B2]	[D5]

Round III

SIFT...

Player # and [Ranking]	New [Ranking]
G 1[D1]	[G1]
2[D2]	[G2]
3[D3]	[G5]
4[D4]	[G4]
5[E1]	[G3]

Round IV

SIFT...

(Same as Round III)

Player #	New [Ranking]
B B1	Becomes [B1]
B2	Becomes [B5]
B3	Becomes [B3]
B4	Becomes [B4]
B5	Becomes [B2]

Player # and [Ranking]	New [Ranking]
E 1[C2]	[E2]
2[A3]	[E1]
3[B3]	[E4]
4[C3]	[E5]
5[A4]	[E3]

Player # and [Ranking]	New [Ranking]
H 1[D5]	[H1]
2[E2]	[H2]
3[E3]	[H4]
4[E4]	[H3]
5[F1]	[H5]

(Same as Round III)

Player #	New [Ranking]
C C1	Becomes [C4]
C2	Becomes [C5]
C3	Becomes [C1]
C4	Becomes [C3]
C5	Becomes [C2]

Player # and [Ranking]	New [Ranking]
F 1[B4]	[F1]
2[C4]	[F3]
3[A5]	[F2]
4[B5]	[F5]
5[C5]	[F4]

Player # and [Ranking]	New [Ranking]
I 1[E5]	[I2]
2[F2]	[I1]
3[F3]	[I4]
4[F4]	[I3]
5[F5]	[I5]

(Same as Round III)

START... SORT...

SIFT ...

SIFT...

SIFT...

LINEAR

DOUBLES AND TRIPLES RANKING TOURNAMENTS

TOURNAMENT EXAMPLE

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Group: P.E. Class Activity: Volleyball

By tracking individuals in the example provided below you can readily see how an ongoing ranking process can be done for a class, team, or any group. On Monday players were placed in groups with one of the "best" players in each: then the #1's were "sorted" up, followed by the #2's, #3's, and so on. Thereafter, "sifting" (a ladder-like process) occurred. Follow "Miller" (the best player in the entire class) who happened to be absent on the first day and see how he earns his way to the top. The "notes" below each column refer to the column above it or the next day.

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	MONDAY
<u>"A" (5)</u> 1. Merritt (1) 2. Vaughn (2) 3. James (3) 4. O'Connor (4) 5. Hall (5) <u>"B" (5)</u> 6. Luinstra (1) 7. Fetcho (4) 8. Amarel (3) 9. Pilkington (5) 10. Smith (2) <u>"C" (5)</u> 11. Sargent (1) 12. Clayton (5) 13. Saad (3) 14. Petrie (4) 15. Ames (2) <u>"D" (5)</u> 16. Butzlaff (2) 17. Ward (5) 18. Herbst (3) 19. Judah (4) 20. Bartlett (1) <u>"E" (7)</u> 21. Pierson (3) 22. Kinder (2) 23. Gibson (7) 24. Demus (1) 25. Flanigan (5) 26. Sicilia (6) 27. Steeve (4) 28. Miller (AB)	<u>"A" (4)</u> 1. Merritt (1) 2. Luinstra (2) 3. Sargent (3) 4. Bartlett (4) <u>"B" (4)</u> 5. Demus (1) 6. Kinder (2) 7. Vaughn (4) 8. Smith (3) <u>"C" (4)</u> 9. Ames (3) 10. Butzlaff (1) 11. Pierson (4) 12. James (2) <u>"D" (4)</u> 13. Amarel (1) 14. Saad (3) 15. Herbst (4) 16. Steeve (2) <u>"E" (5)</u> 17. O'Connor (5) 18. Fetcho (2) 19. Petrie (3) 20. Judah (4) 21. Flanigan (1) <u>"F" (7)</u> 22. Hall (5) 23. Pilkington (4) 24. Clayton (3) 25. Ward (2) 26. Sicilia (AB) 27. Gibson (AB) 28. Miller (1)	<u>"A" (4)</u> 1. Merritt (2) 2. Luinstra (1) 3. Sargent (4) 4. Demus (3) <u>"B" (7)</u> 5. Bartlett (2) 6. Kinder (1) 7. Smith (3) 8. Butzlaff (6) 9. Vaughn (4) 10. James (5) 11. Ames (7) <u>"C" (7)</u> 12. Amarel (AB) 13. Pierson (4) 14. Steeve (3) 15. Saad (5) 16. Flanigan (AB) 17. Herbst (6) 18. Fetcho (1) 19. Petrie (2) 20. Judah (7) <u>"D" (7)</u> 21. Miller (1) 22. O'Connor (7) 23. Ward (2) 24. Clayton (3) 25. Pilkington (6) 26. Hall (5) 27. Sicilia (4) 28. Gibson (AB)	<u>"A" (4)</u> 1. Luinstra (2) 2. Merritt (1) 3. Demus (4) 4. Kinder (3) <u>"B" (4)</u> 5. Sargent (1) 6. Bartlett (2) 7. Smith (3) 8. Vaughn (4) <u>"C" (5)</u> 9. James (1) 10. Fetcho (4) 11. Petrie (5) 12. Butzlaff (3) 13. Ames (2) <u>"D" (5)</u> 14. Steeve (3) 15. Pierson (4) 16. Saad (5) 17. Miller (1) 18. Ward (2) <u>"E" (5)</u> 19. Herbst (5) 20. Judah (4) 21. Amarel (1) 22. Flanigan (2) 23. Clayton (3) <u>"F" (5)</u> 24. Sicilia (1) 25. Hall (3) 26. Pilkington (4) 27. O'Connor (5) 28. Gibson (2)	<u>"A" (7)</u> 1. Merritt (1) 2. Luinstra (2) 3. Kinder (3) 4. Sargent (4) 5. Demus (5) 6. Bartlett (7) 7. Smith (6) <u>"B" (7)</u> 8. James (2) 9. Vaughn (3) 10. Ames (5) 11. Butzlaff (4) 12. Fetcho (6) 13. Miller (1) 14. Petrie (7) <u>"C" (7)</u> 15. Ward (2) 16. Steeve (5) 17. Pierson (4) 18. Amarel (1) 19. Saad (7) 20. Flanigan (3) 21. Clayton (6) <u>"D" (7)</u> 22. Judah (2) 23. Sicilia (1) 24. Herbst (4) 25. Gibson (3) 26. Hall (5) 27. Pilkington (6) 28. O'Connor (7)	<u>"A" (7)</u> 1. Merritt (2) 2. Luinstra (3) 3. Kinder (4) 4. Sargent (5) 5. Demus (6) 6. Miller (1) 7. James (7) <u>"B" (7)</u> 8. Smith (1) 9. Bartlett (2) 10. Vaughn (3) 11. Butzlaff (4) 12. Ames (5) 13. Amarel (7) 14. Ward (6) <u>"C" (7)</u> 15. Fetcho (3) 16. Petrie (2) 17. Flanigan (1) 18. Pierson (4) 19. Steeve (5) 20. Sicilia (6) 21. Judah (7) <u>"D" (7)</u> 22. Clayton (1) 23. Saad (1) 24. Gibson (AB) 25. Herbst (AB) 26. Hall (3) 27. Pilkington (4) 28. O'Connor (5)

Notes:

I. One absent (AB) makes 27: five groups (on narrow courts) of 5, 5, 5, 5, 7.
II. These are the PAIRINGS for Monday's competitions; the RANKINGS from the day's competition are in parenthesis after each name.
III. One player from each group will now go "up" or "down" for the next day (two for the group of seven).
IV. Any player who is absent automatically drops down one group each day.

Notes:

I. Two absent (AB) makes 26: six groups (on narrow courts) of 4, 4, 4, 4, 5, 5.
II. These are the PAIRINGS for Tuesday's competitions; the resulting RANKINGS are in parenthesis after each name.
III. One player from each group will now go "up" or "down" for Wednesday.
IV. Any player who is absent automatically drops down one group each day.

Notes:

I. Three absent (AB) makes 25: four groups (on narrow courts) of 4, 7, 7, 7... Gym use conflict today allows for only two full courts.
II. These are the PAIRINGS for Wednesday's competitions; the resulting RANKINGS are in parenthesis after each name.
III. Two players from each group will now go "up" or "down" for Thursday.
IV. Any player who is absent automatically drops down one group each day.

Notes:

I. No one absent - 28 players: six groups (on narrow courts) of 4, 4, 5, 5, 5, 5.
II. These are the PAIRINGS for today's competitions; the resulting RANKINGS are in parenthesis after each name.
III. One player from each group will now go "up" or "down" for tomorrow.

Notes:

I. No one absent - 28 players: four groups (on narrow courts) of 7, 7, 7, 7.
II. These are the PAIRINGS for today's competitions; the resulting RANKINGS are in parenthesis after each name.
III. Two players from each group will now go "up" or "down" for tomorrow.

SUB-SET COMBINATIONS TO MATCH GROUP SIZES

To use this "cheat-sheet" just do the following: 1) As your group comes in, take attendance and determine the total number of players; 2). While the players are warming-up refer to the "Total in Group" listed below and, based on the number of courts available, select "sub-sets" to match; 3). fill in the names of the players on the appropriate score sheets and; 4) begin play. The entire process takes only a few minutes and players simply divide the amount of time available by the contests to play as shown on their score sheet, and run their own closed-cell/sub-set tourney!

Total in Group	Bjerring Sub-sets	Total in Group	Bjerring Sub-sets	Total in Group	Bjerring Sub-sets
4	4 (Doubles)	24	4, 4, 4, 4, 4, 4 4, 5, 5, 5, 5	39	4, 5, 5, 5, 5, 5, 5, 5 4, 7, 7, 7, 7, 7
5	5 (Doubles)		5, 5, 7, 7	40	5, 5, 5, 5, 5, 5, 5, 5
6	5 (With two players as one number)	25	4, 4, 4, 4, 4, 5 5, 5, 5, 5, 5	41	4, 5, 5, 5, 5, 5, 5, 7 4, 4, 4, 4, 4, 4, 4, 4, 5
7	7 (Triples)		4, 7, 7, 7		4, 4, 4, 4, 4, 7, 7, 7
8	4, 4	26	4, 4, 4, 4, 5, 5 4, 4, 4, 7, 7, 5, 7, 7, 7,	42	4, 4, 4, 4, 4, 4, 4, 4, 5, 5 4, 4, 4, 4, 4, 4, 4, 7, 7 4, 4, 4, 5, 5, 5, 5, 5
9	4, 5				
10	5, 5	27	4, 4, 4, 5, 5, 5 4, 4, 4, 7, 7 5, 5, 5, 5, 7	43	4, 4, 4, 4, 4, 4, 4, 5, 5, 5 4, 4, 4, 4, 4, 4, 5, 7, 7 4, 4, 4, 4, 5, 5, 5, 5 4, 4, 4, 5, 5, 7, 7, 7 4, 4, 7, 7, 7, 7, 7,
11	4, 7	28	4, 4, 5, 5, 5, 5 4, 5, 5, 7, 7		
12	4, 4, 4				
13	4, 4, 5	29	4, 5, 5, 5, 5, 5 4, 4, 7, 7, 7	44	4, 4, 4, 4, 4, 4, 5, 5, 5, 5 4, 4, 4, 4, 4, 5, 5, 7, 7 4, 4, 4, 5, 5, 5, 5, 5, 7 4, 4, 5, 5, 5, 7, 7, 7 4, 5, 7, 7, 7, 7, 7
14	4, 5, 5	30	4, 4, 4, 4, 4, 5, 5 5, 5, 5, 5, 5, 5 7, 7, 7, 5, 4	45	4, 4, 4, 4, 4, 5, 5, 5, 5, 5 4, 4, 4, 4, 5, 5, 5, 7, 7 4, 4, 4, 5, 7, 7, 7, 7
15	5, 5, 5				
16	4, 4, 4, 4 4, 5, 7	31	4, 4, 4, 4, 5, 5, 5 4, 4, 4, 4, 4, 4, 7	46	4, 4, 4, 4, 5, 5, 5, 5, 5, 5 4, 4, 4, 5, 5, 5, 5, 7, 7 4, 4, 5, 5, 7, 7, 7, 7
17	4, 4, 4, 5 5, 5, 7	32	4, 4, 4, 4, 4, 4, 4, 4 5, 5, 5, 5, 5, 5, 7 7, 7, 7, 7, 7, 4	47	4, 4, 4, 5, 5, 5, 5, 5, 5, 5 4, 4, 4, 4, 5, 5, 7, 7, 7 4, 4, 5, 5, 5, 5, 5, 7, 7 4, 4, 4, 7, 7, 7, 7, 7
18	4, 4, 5, 5 4, 7, 7				
19	5, 5, 5, 4 7, 7, 5	33	4, 4, 4, 4, 4, 4, 4, 5 7, 7, 7, 7, 5	48	4, 4, 5, 5, 5, 5, 5, 5, 5, 5 4, 4, 4, 5, 5, 5, 7, 7, 7 4, 5, 5, 5, 5, 5, 5, 7, 7 4, 4, 5, 7, 7, 7, 7, 7
20	4, 4, 4, 4, 4, 4, 4, 5, 7 5, 5, 5, 5	34	4, 4, 4, 4, 4, 4, 5, 5		
21	4, 4, 4, 4, 5 4, 5, 5, 7 7, 7, 7	35	5, 5, 5, 5, 5, 5, 5, 5, 7, 7, 7, 7, 7	49	4, 5, 5, 5, 5, 5, 5, 5, 5, 5 4, 4, 5, 5, 5, 5, 7, 7, 7 5, 5, 5, 5, 5, 5, 5, 7, 7 4, 5, 5, 7, 7, 7, 7, 7
22	4, 4, 4, 5, 5 4, 4, 7, 7 5, 5, 5, 7	36	4, 5, 5, 5, 5, 5, 7 4, 4, 7, 7, 7		
23	4, 4, 5, 5, 5 4, 5, 7, 7	37	5, 5, 5, 5, 5, 5, 7 4, 5, 7, 7, 7	50	4, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5 4, 4, 4, 4, 5, 5, 5, 5, 7, 7 4, 4, 4, 5, 5, 5, 7, 7, 7
		38	4, 4, 5, 5, 5, 5, 5, 5, 5, 5, 7, 7, 7, 7		

TOURNAMENT PAIRINGS SCHEDULES

Four Players

Court 1

Time 1	1, 2 V 3, 4
Time 2	1, 3 V 2, 4
Time 3	1, 4 v 2, 3

Five Players

Court 1

Time 1	1, 2 V 3, 4
Time 2	1, 3 V 2, 5
Time 3	1, 4 V 3, 5
Time 4	1, 5 V 2, 4
Time 5	2, 3 V 4, 5

Eight Players

Court A

Time 1	1, 2 V 3, 4
Time 2	1, 5 V 2, 6
Time 3	1, 7 V 2, 8
Time 4	1, 3 V 5, 7
Time 5	1, 8 V 3, 6
Time 6	1, 4 V 5, 8
Time 7	1, 6 V 4, 7

Court B

5, 6 V 7, 8
3, 7 V 4, 8
3, 5 V 4, 6
2, 4 V 6, 8
2, 7 V 4, 5
2, 3 V 6, 7
2, 5 V 3, 8

Twelve Players

(Two Courts)

Court 1

Time 1	1, 2 V 6, 7
Time 2	5, 9 V 8, 11
Time 3	1, 3 V 7, 8
Time 4	6, 10 V 9, 12
Time 5	1, 4 V 8, 9
Time 6	7, 11 V 2, 10
Time 7	1, 5 V 9, 10
Time 8	1, 6 V 10, 11
Time 9	1, 7 V 11, 12
Time 10	1, 8 V 2, 12
Time 11	4, 11 V 3, 6
Time 12	1, 9 V 2, 3
Time 13	5, 12 V 4, 7
Time 14	1, 10 V 3, 4
Time 15	1, 11 V 4, 5
Time 16	1, 12 V 5, 6
Time 17	2, 6 V 5, 8

Court 2

3, 12 V 4, 10
4, 6 V 2, 7
2, 4 V 5, 11
5, 7 V 3, 8
3, 5 V 6, 12
6, 8 V 4, 9
8, 12 V 3, 11
2, 9 V 4, 12
3, 10 V 2, 5
7, 9 V 5, 10
10, 12 V 2, 8
8, 10 V 6, 11
2, 11 V 3, 9
9, 11 V 7, 12
3, 7 V 6, 9
4, 8 V 7, 10

TWELVE PLAYERS

(On Three Courts)

GAME	COURT I	COURT II	COURT III
A	1, 2 V 6, 7	3, 12 V 4, 10	5, 9 V 8, 11
B	1, 3 V 7, 8	2, 4 V 5, 11	6, 10 V 9, 12
C	1, 4 V 8, 9	3, 5 V 6, 12	7, 11 V 2, 10
D	1, 5 V 9, 10	4, 6 V 2, 7	8, 12 V 3, 11
E	1, 6 V 10, 11	5, 7 V 3, 8	2, 9 V 4, 12
F	1, 7 V 11, 12	6, 8 V 4, 9	3, 10 V 2, 5
G	1, 8 V 2, 12	7, 9 V 5, 10	4, 11 V 3, 6
H	1, 9 V 2, 3	8, 10 V 6, 11	5, 12 V 4, 7
I	1, 10 V 3, 4	9, 11 V 7, 12	2, 6 V 5, 8
J	1, 11 V 4, 5	10, 12 V 2, 8	3, 7 V 6, 9
K	1, 12 V 5, 6	2, 11 V 3, 9	4, 8 V 7, 10

SIXTEEN PLAYERS

(On Four Courts)

GAME	COURT I	COURT II	COURT III	COURT IV
A	1, 2 V 7, 12	3, 4 V 6, 10	5, 13 V 14, 16	8, 9 V 11, 15
B	1, 7 V 2, 12	3, 6 V 4, 10	5, 14 V 13, 16	8, 11 V 9, 15
C	1, 12 V 2, 7	3, 10 V 4, 6	5, 16 V 13, 14	8, 15 V 9, 11
D	1, 3 V 8, 13	4, 5 V 7, 11	2, 6 V 14, 15	9, 10 V 12, 16
E	1, 8 V 3, 13	4, 7 V 5, 11	2, 14 V 6, 15	9, 12 V 10, 16
F	1, 13 V 3, 8	4, 11 V 5, 7	2, 15 V 6, 14	9, 16 V 10, 12
G	1, 4 V 9, 14	5, 6 V 8, 12	3, 7 V 15, 16	2, 10 V 11, 13
H	1, 9 V 4, 14	5, 8 V 6, 12	3, 15 V 7, 16	2, 11 V 10, 13
I	1, 14 V 4, 9	5, 12 V 6, 8	3, 16 V 7, 15	2, 13 V 10, 11
J	1, 5 V 10, 15	6, 7 V 9, 13	2, 4 V 8, 16	3, 11 V 12, 14
K	1, 10 V 5, 15	6, 9 V 7, 13	2, 8 V 4, 16	3, 12 V 11, 14
L	1, 15 V 5, 10	6, 13 V 7, 9	2, 16 V 4, 8	3, 14 V 11, 12
M	1, 6 V 11, 16	7, 8 V 10, 14	2, 3 V 5, 9	4, 12 V 13, 15
N	1, 11 V 6, 16	7, 10 V 8, 14	2, 5 V 3, 9	4, 13 V 12, 15
O	1, 16 V 6, 11	7, 14 V 8, 10	2, 9 V 3, 5	4, 15 V 12, 13

Twenty - Five Players

(On Five Courts)

Game	Court A	Court B	Court C	Court D	Court E
A	1, 2 V 3, 4	6, 7 V 8, 9	11, 12 V 13, 14	16, 17 V 18, 19	21, 22 V 23, 24
B	1, 3 V 2, 5	6, 8 V 7, 10	11, 13 V 12, 15	16, 18 V 17, 20	21, 23 V 22, 25
C	1, 4 V 3, 5	6, 9 V 8, 10	11, 14 V 13, 15	16, 19 V 18, 20	21, 24 V 23, 25
D	1, 5 V 2, 4	6, 10 V 7, 9	11, 15 V 12, 14	16, 20 V 17, 19	21, 25 V 22, 24
E	2, 3 V 4, 5	7, 8 V 9, 10	12, 13 V 14, 15	17, 18 V 19, 20	22, 23 V 24, 25
F	1, 6 V 11, 16	2, 7 V 13, 19	3, 8 V 15, 17	4, 9 V 12, 20	5, 10 V 14, 18
G	1, 11 V 6, 21	2, 13 V 7, 25	3, 15 V 8, 24	4, 12 V 9, 23	5, 14 V 10, 22
H	1, 16 V 11, 21	2, 19 V 13, 25	3, 17 V 15, 24	4, 20 V 12, 23	5, 18 V 14, 22
I	1, 21 V 6, 16	2, 25 V 7, 19	3, 24 V 8, 17	4, 23 V 9, 20	5, 22 V 10, 18
J	6, 11 V 16, 21	7, 13 V 19, 25	8, 15 V 17, 24	9, 12 V 20, 23	10, 14 V 18, 22
K	1, 10 V 15, 19	2, 6 V 12, 17	3, 7 V 14, 20	4, 8 V 11, 18	5, 9 V 13, 16
L	1, 15 V 10, 23	2, 12 V 6, 22	3, 14 V 7, 21	4, 11 V 8, 25	5, 13 V 9, 24
M	1, 19 V 15, 23	2, 17 V 12, 22	3, 20 V 14, 21	4, 18 V 11, 25	5, 16 V 13, 24
N	1, 23 V 10, 19	2, 22 V 6, 17	3, 21 V 7, 20	4, 25 V 8, 18	5, 24 V 9, 16
O	10, 15 V 19, 23	6, 12 V 17, 22	7, 14 V 20, 21	8, 11 V 18, 25	9, 13 V 16, 24
P	1, 9 V 14, 17	2, 10 V 11, 20	3, 6 V 13, 18	4, 7 V 15, 16	5, 8 V 12, 19
Q	1, 14 V 9, 25	2, 11 V 10, 24	3, 13 V 6, 23	4, 15 V 7, 22	5, 12 V 8, 21
R	1, 17 V 14, 25	2, 20 V 11, 24	3, 18 V 13, 23	4, 16 V 15, 22	5, 19 V 12, 21
S	1, 25 V 9, 17	2, 24 V 10, 20	3, 23 V 6, 18	4, 22 V 7, 16	5, 21 V 8, 19
T	9, 14 V 17, 25	10, 11 V 20, 24	6, 13 V 18, 23	7, 15 V 16, 22	8, 12 V 19, 21
U	1, 8 V 13, 20	2, 9 V 15, 18	3, 10 V 12, 16	4, 6 V 14, 19	5, 7 V 11, 17
V	1, 13 V 8, 22	2, 15 V 9, 21	3, 12 V 10, 25	4, 14 V 6, 24	5, 11 V 7, 23
W	1, 20 V 13, 22	2, 18 V 15, 21	3, 16 V 12, 25	4, 19 V 14, 24	5, 17 V 11, 23
X	1, 22 V 8, 20	2, 21 V 9, 18	3, 25 V 10, 16	4, 24 V 6, 19	5, 23 V 7, 17
Y	8, 13 V 20, 22	9, 15 V 18, 21	10, 12 V 16, 25	6, 14 V 19, 24	7, 11 V 17, 23
Z	1, 7 V 12, 18	2, 8 V 14, 16	3, 9 V 11, 19	4, 10 V 13, 17	5, 6 V 15, 20
AA	1, 12 V 7, 24	2, 14 V 8, 23	3, 11 V 9, 22	4, 13 V 10, 21	5, 15 V 6, 25
BB	1, 18 V 12, 24	2, 16 V 14, 23	3, 19 V 11, 22	4, 17 V 13, 21	5, 20 V 15, 25
CC	1, 24 V 7, 18	2, 23 V 8, 16	3, 22 V 9, 19	4, 21 V 10, 17	5, 25 V 6, 20
DD	7, 12 V 18, 24	8, 14 V 16, 23	9, 11 V 19, 22	10, 13 V 17, 21	6, 15 V 20, 25

TRIPLES

Every player plays with every other player twice and against every other player three times. Use a court of your own choice but it is recommended that a 4.5 m by 18 m court be used. With the two systems listed below you can run 7, 9, 14, 16, 18, etc. by using combinations. The "Out" player officiates and supplies volleyballs.

SEVEN PLAYERS

(Triples On One Court)

GAME

A	1,	2,	3,	V	4,	5,	6
B	1,	4,	7	V	2,	3,	5
C	2,	6,	7	V	1,	3,	4
D	2,	4,	5	V	3,	6,	7
E	1,	5,	7	V	3,	4,	6
F	1,	2,	6	V	3,	5,	7
G	1,	5,	6	V	2,	4,	7

NINE PLAYERS

(Triples On One Court)

GAME

A	1,	4,	7	V	2,	5,	9
B	1,	5,	8	V	2,	6,	7
C	1,	6,	9	V	2,	4,	8
D	2,	4,	8	V	3,	5,	7
E	2,	5,	9	V	3,	6,	8
F	2,	6,	7	V	3,	4,	9
G	3,	4,	9	V	1,	5,	8
H	3,	5,	7	V	1,	6,	9
I	3,	6,	8	V	1,	4,	7
J	1,	2,	3	V	4,	5,	6
K	4,	5,	6	V	7,	8,	9
L	7,	8,	9	V	1,	2,	3

**DOUBLES AND TRIPLES
RANKING TOURNAMENTS**

(The following is a personal correspondence from James Bjerring, March 22, 1990)

Dear Mike:

Re: Balanced Doubles and Triples Designs

I've discovered a new class of doubles and triples designs. That's the good news. The bad news is that in most cases only one court can be used at a time (because in these cases each game or subtournament contains at least one player from every other game or subtournament. That's ok when the number of players is low. When there's a lot of players, however, such designs might be most useful for a tournament played over several sessions or even a season.

Another new feature in these designs is that some of them contain more than one rep! For example, in a doubles tournament with two reps, each player would play with each other player as a partner twice, and as an opponent four times, though with a different mix of players in each rep. In any event, the designs are as follows:

Doubles: 6 Players/

1 Court

1, 2 v 3, 4
1, 3 v 2, 5
1, 4 v 3, 5
1, 5 v 2, 4
2, 3 v 4, 5
1, 2 v 3, 4
1, 3 v 2, 6
1, 4 v 3, 6
1, 6 v 2, 4
2, 3 v 4, 6
1, 2 v 3, 5
1, 3 v 2, 6
1, 5 v 3, 6
1, 6 v 2, 5
2, 3 v 5, 6
1, 2 v 4, 5
1, 4 v 2, 6
1, 5 v 4, 6
1, 6 v 2, 5
2, 4 v 5, 6
1, 3 v 4, 5
1, 4 v 3, 6
1, 5 v 4, 6
1, 6 v 3, 5
3, 4 v 5, 6
2, 3 v 4, 5
2, 4 v 3, 6
2, 5 v 4, 6
2, 6 v 3, 5
3, 4 v 5, 6

Doubles: 7 Players/

1 Court

1, 3 v 4, 5
1, 4 v 3, 5
1, 5 v 3, 4
2, 4 v 5, 6
2, 5 v 4, 6
2, 6 v 4, 5
3, 5 v 6, 7
3, 6 v 5, 7
3, 7 v 5, 6
4, 6 v 7, 1
4, 7 v 6, 1
4, 1 v 6, 7
5, 7 v 1, 2
5, 1 v 7, 2
5, 2 v 7, 1
6, 1 v 2, 3
6, 2 v 1, 3
6, 3 v 1, 2
7, 2 v 3, 4
7, 3 v 2, 4
7, 4 v 2, 3

LINEAR**DOUBLES AND TRIPLES
RANKING TOURNAMENTS**

13

Doubles - 9 Players**3 Reps, One Court**

1, 2 V 3, 5
1, 3 V 2, 5
1, 5 V 2, 3
2, 3 V 4, 6
2, 4 V 3, 6
2, 6 V 3, 4
3, 4 V 5, 7
3, 5 V 4, 7
3, 7 V 4, 5
4, 5 V 6, 8
4, 6 V 5, 8
4, 8 V 5, 6
5, 6 V 7, 9
5, 7 V 6, 9
5, 9 V 6, 7
6, 7 V 8, 1
6, 8 V 7, 1
6, 1 V 7, 8
7, 8 V 9, 2
7, 9 V 8, 2
7, 2 V 8, 9
8, 9 V 1, 3
8, 1 V 9, 3
8, 3 V 9, 1
9, 1 V 2, 4
9, 2 V 1, 4
9, 4 V 1, 2
1, 4 V 6, 9
1, 6 V 4, 9
1, 9 V 4, 6
2, 5 V 7, 1
2, 7 V 5, 1
2, 1 V 5, 7
3, 6 V 8, 2
3, 8 V 6, 2
3, 2 V 6, 8
4, 7 V 9, 3
4, 9 V 7, 3
4, 3 V 7, 9
5, 8 V 1, 4
5, 1 V 8, 4
5, 4 V 8, 1
6, 9 V 2, 5
6, 2 V 9, 5
6, 5 V 9, 2
7, 1 V 3, 6
7, 3 V 1, 6
7, 6 V 1, 3
8, 2 V 4, 7
8, 4 V 2, 7
8, 7 V 2, 4
9, 3 V 5, 8
9, 5 V 3, 8
9, 8 V 3, 5

Doubles - 10 Players**2 Reps, One Court**

1, 2 V 5, 8
1, 5 V 2, 8
1, 8 V 2, 5
1, 3 V 6, 9
1, 6 V 3, 9
1, 9 V 3, 6
1, 4 V 7, 10
1, 7 V 4, 10
1, 10 V 4, 7
2, 3 V 5, 10
2, 5 V 3, 10
2, 10 V 3, 5
2, 3 V 4, 6
2, 4 V 3, 6
2, 6 V 3, 4
3, 4 V 7, 8
3, 7 V 4, 8
3, 8 V 4, 7
1, 2 V 4, 9
1, 4 V 2, 9
1, 9 V 2, 4
1, 5 V 6, 7
1, 6 V 5, 7
1, 7 V 5, 6
2, 6 V 7, 8
2, 7 V 6, 8
2, 8 V 6, 7
3, 5 V 7, 9
3, 7 V 5, 9
3, 9 V 5, 7
4, 5 V 6, 10
4, 6 V 5, 10
4, 10 V 5, 6
4, 5 V 8, 9
4, 8 V 5, 9
4, 9 V 5, 8
6, 8 V 9, 10
6, 9 V 8, 10
6, 10 V 8, 9
2, 7 V 9, 10
2, 9 V 7, 10
2, 10 V 7, 9
1, 3 V 8, 10
1, 8 V 3, 10
1, 10 V 3, 8

Doubles - 11 Players**2 Reps, One Court**

1, 5 V 6, 7
1, 6 V 5, 9
1, 7 V 6, 9
1, 9 V 5, 7
5, 6 V 7, 9
2, 6 V 7, 8
2, 7 V 6, 10
2, 8 V 7, 10
2, 10 V 6, 8
6, 7 V 8, 10
3, 7 V 8, 9
3, 8 V 7, 11
3, 9 V 8, 11
3, 11 V 7, 9
7, 8 V 9, 11
4, 8 V 9, 10
4, 9 V 8, 1
4, 10 V 9, 1
4, 1 V 8, 10
8, 9 V 10, 1
5, 9 V 10, 11
5, 10 V 9, 2
5, 11 V 10, 2
5, 2 V 9, 11
9, 10 V 11, 2
6, 10 V 11, 1
6, 11 V 10, 3
6, 1 V 11, 3
6, 3 V 10, 1
10, 11 V 1, 3
7, 11 V 1, 2
7, 1 V 11, 4
7, 2 V 1, 4
7, 4 V 11, 2
11, 1 V 2, 4
8, 1 V 2, 3
8, 2 V 1, 5
8, 3 V 2, 5
8, 5 V 1, 3
1, 2 V 3, 5
9, 2 V 3, 4
9, 3 V 2, 6
9, 4 V 3, 6
9, 6 V 2, 4
2, 3 V 4, 6
10, 3 V 4, 5
10, 4 V 3, 7
10, 5 V 4, 7
10, 7 V 3, 5
3, 4 V 5, 7
11, 4 V 5, 6
11, 5 V 4, 8
11, 6 V 5, 8
11, 8 V 4, 6
4, 5 V 6, 8

Doubles - 13 Players**1 Reps, One Court**

1, 2 V 4, 10
1, 4 V 2, 10
1, 10 V 2, 4
2, 3 V 5, 11
2, 5 V 3, 11
2, 11 V 3, 5
3, 4 V 6, 12
3, 6 V 4, 12
3, 12 V 4, 6
4, 5 V 7, 13
4, 7 V 5, 13
4, 13 V 5, 7
5, 6 V 8, 1
5, 8 V 6, 1
5, 1 V 6, 8
6, 7 V 9, 2
6, 9 V 7, 2
6, 2 V 7, 9
7, 8, V 10, 3
7, 10 V 8, 3
7, 3 V 8, 10
8, 9 V 11, 4
8, 11 V 9, 4
8, 4 V 9, 11
9, 10 V 12, 5
9, 12 V 10, 5
9, 5 V 10, 12
10, 11 V 13, 6
10, 13 V 11, 6
10, 6 V 11, 13
11, 12 V 1, 7
11, 1 V 12, 7
11, 7 V 12, 1
12, 13 V 2, 8
12, 2 V 13, 8
12, 8 V 13, 2
13, 1 V 3, 9
13, 3 V 1, 9
13, 9 V 1, 3

Doubles: 15 Players 4 Reps / 2 Courts

	<u>Court I</u>	<u>Court II</u>
A		
1	1, 4 V 5, 7	9, 10 V 11, 12
2	1, 9 V 4, 10	5, 11 V 7, 12
3	1, 11 V 4, 12	5, 9 V 7, 10
4	1, 5 V 9, 11	4, 7 V 10, 12
5	1, 12 V 5, 10	4, 11 V 7, 9
6	1, 7 V 9, 12	4, 5 V 10, 11
7	1, 10 V 7, 11	4, 9 V 5, 12
B		
1	2, 5 V 6, 8	10, 11 V 12, 13
2	2, 10 V 5, 11	6, 12 V 8, 13
3	2, 12 V 5, 13	6, 10 V 8, 11
4	2, 6 V 10, 12	5, 8 V 11, 13
5	2, 13 V 6, 11	5, 12 V 8, 10
6	2, 8 V 10, 13	5, 6 V 11, 12
7	2, 11 V 8, 12	5, 10 V 6, 13
C		
1	3, 6 V 7, 9	11, 12 V 13, 14
2	3, 11 V 6, 12	7, 13 V 9, 14
3	3, 13 V 6, 14	7, 11 V 9, 12
4	3, 7 V 11, 13	6, 9 V 12, 14
5	3, 14 V 7, 12	6, 13 V 9, 11
6	3, 9 V 11, 14	6, 7 V 12, 13
7	3, 12 V 9, 13	6, 11 V 7, 14
D		
1	4, 7 V 8, 10	12, 13 V 14, 15
2	4, 12 V 7, 13	8, 14 V 10, 15
3	4, 14 V 7, 15	8, 12 V 10, 13
4	4, 8 V 12, 14	7, 10 V 13, 15
5	4, 15 V 8, 13	7, 14 V 10, 12
6	4, 10 V 12, 15	7, 8 V 13, 14
7	4, 13 V 10, 15	7, 12 V 8, 15
E		
1	5, 8 V 9, 11	13, 14 V 15, 1
2	5, 13 V 8, 14	9, 15 V 11, 1
3	5, 15 V 8, 1	9, 13 V 11, 14
4	5, 9 V 13, 15	8, 1 V 14, 1
5	5, 1 V 9, 14	8, 15 V 11, 13
6	5, 11 V 13, 1	8, 9 V 14, 15
7	5, 14 V 11, 15	8, 13 V 9, 1
F		
1	6, 9 V 10, 12	14, 15 V 1, 2
2	6, 14 V 9, 14	10, 1 V 12, 2
3	6, 1 V 9, 2	10, 14 V 12, 15
4	6, 10 V 14, 1	9, 12 V 15, 2
5	6, 2 V 10, 15	9, 1 V 12, 14
6	6, 12 V 14, 2	9, 10 V 15, 1
7	6, 15 V 12, 1	9, 14 V 10, 2

	<u>Court I</u>	<u>Court II</u>
G		
1	7, 10 V 11, 13	15, 1 V 2, 3
2	7, 15 V 10, 1	11, 2 V 13, 3
3	7, 2 V 10, 3	11, 15 V 13, 1
4	7, 11 V 15, 2	10, 13 V 1, 3
5	7, 3 V 11, 1	10, 2 V 13, 15
6	7, 13 V 15, 3	10, 11 V 1, 2
7	7, 1 V 13, 2	10, 15 V 11, 3
H		
1	8, 11 V 12, 14	1, 2 V 3, 4
2	8, 1 V 11, 2	12, 3 V 14, 4
3	8, 3 V 11, 4	12, 1 V 14, 2
4	8, 12 V 1, 3	11, 14 V 2, 4
5	8, 4 V 12, 2	11, 3 V 14, 1
6	8, 14 V 1, 4	11, 12 V 2, 3
7	8, 2 V 14, 3	11, 1 V 12, 4
I		
1	9, 12 V 13, 15	2, 3 V 4, 5
2	9, 2 V 12, 3	13, 4 V 15, 5
3	9, 4 V 12, 5	13, 2 V 15, 3
4	9, 13 V 2, 4	12, 15 V 3, 5
5	9, 5 V 13, 3	12, 4 V 15, 2
6	9, 15 V 2, 5	12, 13 V 3, 4
7	9, 3 V 15, 4	12, 2 V 13, 5
J		
1	10, 13 V 14, 1	3, 4 V 5, 6
2	10, 3 V 13, 4	14, 5 V 1, 6
3	10, 5 V 13, 6	14, 3 V 1, 4
4	10, 14 V 3, 5	13, 1 V 4, 6
5	10, 6 V 14, 4	13, 5 V 1, 3
6	10, 1 V 3, 6	13, 14 V 4, 5
7	10, 4 V 1, 5	13, 3 V 14, 6
K		
1	11, 14 V 15, 2	4, 5 V 6, 7
2	11, 4 V 14, 5	15, 6 V 2, 7
3	11, 6 V 14, 7	15, 4 V 2, 5
4	11, 15 V 4, 6	14, 2 V 5, 7
5	11, 7 V 15, 5	14, 6 V 2, 4
6	11, 2 V 4, 7	14, 15 V 5, 6
7	11, 5 V 2, 6	14, 4 V 15, 7
L		
1	12, 15 V 1, 3	5, 6 V 7, 8
2	12, 5 V 15, 6	1, 7 V 3, 8
3	12, 7 V 15, 8	1, 5 V 3, 6
4	12, 1 V 5, 7	15, 3 V 6, 8
5	12, 8 V 1, 6	15, 7 V 3, 5
6	12, 3 V 5, 8	15, 1 V 6, 7
7	12, 6 V 3, 7	15, 5 V 1, 8

	<u>Court I</u>	<u>Court II</u>
M		
1	13, 1 V 2, 4	6, 7 V 8, 9
2	13, 6 V 1, 7	2, 8 V 4, 9
3	13, 8 V 1, 9	2, 6 V 4, 7
4	13, 2 V 6, 8	1, 4 V 7, 9
5	13, 9 V 2, 7	1, 8 V 4, 6
6	13, 4 V 6, 9	1, 2 V 7, 8
7	13, 7 V 4, 8	1, 6 V 2, 9
N		
1	14, 2 V 3, 5	7, 8 V 9, 10
2	14, 7 V 2, 8	3, 9 V 5, 10
3	14, 9 V 2, 10	3, 7 V 5, 8
4	14, 3 V 7, 9	2, 5 V 8, 10
5	14, 10 V 3, 8	2, 9 V 5, 7
6	14, 5 V 7, 10	2, 3 V 8, 9
7	14, 8 V 5, 9	2, 7 V 3, 10
O		
1	15, 3 V 4, 6	8, 9 V 10, 11
2	15, 8 V 3, 9	4, 10 V 6, 11
3	15, 10 V 3, 11	4, 8 V 6, 9
4	15, 4 V 8, 10	3, 6 V 9, 11
5	15, 11 V 4, 9	3, 10 V 6, 8
6	15, 6 V 8, 11	3, 4 V 9, 10
7	15, 9 V 6, 10	3, 8 V 4, 11

Doubles: 21 Players **1 Rep / 1 Court**

A

1. 1, 4 V 5, 10
2. 1, 5 V 4, 12
3. 1, 10 V 5, 12
4. 1, 12 V 4, 10
5. 4, 5 V 10, 12

B

1. 2, 5 V 6, 11
2. 2, 6 V 5, 13
3. 2, 11 V 6, 13
4. 2, 13 V 5, 11
5. 5, 6 V 11, 13

C

1. 3, 6 V 7, 12
2. 3, 7 V 6, 14
3. 3, 12 V 7, 14
4. 3, 14 V 6, 12
5. 6, 7 V 12, 14

D

1. 4, 7 V 8, 13
2. 4, 8 V 7, 15
3. 4, 13 V 8, 15
4. 4, 15 V 7, 13
5. 7, 8 V 13, 15

E

1. 5, 8 V 9, 14
2. 5, 9 V 8, 16
3. 5, 14 V 9, 16
4. 5, 16 V 8, 14
5. 8, 9 V 14, 16

F

1. 6, 9 V 10, 15
2. 6, 10 V 9, 17
3. 6, 15 V 10, 17
4. 6, 17 V 9, 15
5. 9, 10 V 15, 17

G

1. 7, 10 V 11, 16
2. 7, 11 V 10, 18
3. 7, 16 V 11, 18
4. 7, 18 V 10, 16
5. 10, 11 V 16, 18

H

1. 8, 11 V 12, 17
2. 8, 12 V 11, 19
3. 8, 17 V 12, 19
4. 8, 19 V 11, 17
5. 11, 12 V 17, 19

I

1. 9, 12 V 13, 18
2. 9, 13 V 12, 20
3. 9, 18 V 13, 20
4. 9, 20 V 12, 18
5. 12, 13 V 18, 20

J

1. 10, 13 V 14, 19
2. 10, 14 V 13, 21
3. 10, 19 V 14, 21
4. 10, 21 V 13, 19
5. 13, 14 V 19, 21

K

1. 11, 14 V 15, 20
2. 11, 15 V 14, 1
3. 11, 20 V 15, 1
4. 11, 1 V 14, 20
5. 14, 15 V 20, 1

L

1. 12, 15 V 16, 21
2. 12, 16 V 15, 12
3. 12, 21 V 16, 2
4. 12, 2 V 15, 21
5. 15, 16 V 21, 2

M

1. 13, 16 V 17, 1
2. 13, 17 V 16, 13
3. 13, 1 V 17, 3
4. 13, 3 V 16, 1
5. 16, 17 V 1, 3

N

1. 14, 17 V 18, 2
2. 14, 18 V 17, 4
3. 14, 2 V 18, 4
4. 14, 4 V 17, 2
5. 17, 18 V 2, 4

O

1. 15, 18 V 19, 3
2. 15, 19 V 18, 5
3. 15, 3 V 19, 5
4. 15, 5 V 18, 3
5. 18, 19 V 3, 5

P

1. 16, 19 V 20, 4
2. 16, 20 V 19, 6
3. 16, 4 V 20, 6
4. 16, 6 V 19, 4
5. 19, 20 V 4, 6

Q

1. 16, 19 V 20, 4
2. 16, 20 V 19, 6
3. 16, 4 V 20, 6
4. 16, 6 V 19, 4
5. 19, 20 V 4, 6

R

1. 17, 20 V 21, 5
2. 17, 21 V 20, 7
3. 17, 5 V 21, 7
4. 17, 7 V 20, 5
5. 20, 21 V 5, 7

S

1. 18, 21 V 1, 6
2. 18, 1 V 21, 8
3. 18, 6 V 1, 8
4. 18, 8 V 21, 6
5. 21, 1 V 6, 8

T

1. 19, 1 V 2, 7
2. 19, 2 V 1, 9
3. 19, 7 V 2, 9
4. 19, 9 V 1, 7
5. 1, 2 V 7, 9

U

1. 20, 2 V 3, 8
2. 20, 3 V 2, 10
3. 20, 8 V 3, 8
4. 20, 10 V 2, 8
5. 2, 3 V 8, 10

V

1. 21, 3 V 4, 9
2. 21, 4 V 3, 11
3. 21, 9 V 4, 11
4. 21, 11 V 3, 9
5. 3, 4 V 9, 11

Triples: 8 Players 6 Reps / 1 Court

A

- 1, 2, 3 V 4, 5, 6
- 1, 4, 7 V 2, 3, 5
- 2, 6, 7 V 1, 3, 4
- 2, 4, 5 V 3, 6, 7
- 1, 5, 7 V 3, 4, 6
- 1, 2, 6 V 3, 5, 7
- 1, 5, 6 V 2, 4, 7

B

- 1, 2, 3 V 4, 5, 6
- 1, 4, 8 V 2, 3, 5
- 2, 6, 8 V 1, 3, 4
- 2, 4, 5 V 3, 6, 8
- 1, 5, 8 V 3, 4, 6
- 1, 2, 6 V 3, 5, 8
- 1, 5, 6 V 2, 4, 8

C

- 1, 2, 3 V 4, 5, 7
- 1, 4, 8 V 2, 3, 5
- 2, 7, 8 V 1, 3, 4
- 2, 4, 5 V 3, 7, 8
- 1, 5, 8 V 3, 4, 7
- 1, 2, 7 V 3, 5, 8
- 1, 5, 7 V 2, 4, 8

D

- 1, 2, 3 V 4, 6, 7
- 1, 4, 8 V 2, 3, 6
- 2, 7, 8 V 1, 3, 4
- 2, 4, 6 V 3, 7, 8
- 1, 6, 8 V 3, 4, 7
- 1, 2, 7 V 3, 6, 8
- 1, 6, 7 V 2, 4, 8

E

- 1, 2, 3 V 5, 6, 7
- 1, 5, 8 V 2, 3, 6
- 2, 7, 8 V 1, 3, 5
- 2, 5, 6 V 3, 7, 8
- 1, 6, 8 V 3, 5, 7
- 1, 2, 7 V 3, 6, 8
- 1, 6, 7 V 2, 5, 8

F

- 1, 2, 4 V 5, 6, 7
- 1, 5, 8 V 2, 4, 6
- 2, 7, 8 V 1, 4, 5
- 2, 5, 6 V 4, 7, 8
- 1, 6, 8 V 4, 5, 7
- 1, 2, 7 V 4, 6, 8
- 1, 6, 7 V 2, 5, 8

G

- 1, 3, 4 V 5, 6, 7
- 1, 5, 8 V 3, 4, 6
- 3, 7, 8 V 1, 4, 5
- 3, 5, 6 V 4, 7, 8
- 1, 6, 8 V 4, 5, 7
- 1, 3, 7 V 4, 6, 8
- 1, 6, 7 V 3, 5, 8

H

- 2, 3, 4 V 5, 6, 7
- 2, 5, 8 V 3, 4, 6
- 3, 7, 8 V 2, 4, 5
- 3, 5, 6 V 4, 7, 8
- 2, 6, 8 V 4, 5, 7
- 2, 3, 7 V 4, 6, 8
- 2, 6, 7 V 3, 5, 8

Triples: 15 Players 3 Reps / 1 Court

A

- 1, 2, 3 V 5, 6, 9
- 1, 5, 11 V 2, 3, 6
- 2, 9, 11 V 1, 3, 5
- 2, 5, 6 V 3, 9, 11
- 1, 6, 11 V 3, 5, 9
- 1, 2, 9 V 3, 6, 11
- 1, 6, 9 V 2, 5, 11

B

- 2, 3, 4 V 6, 7, 10
- 2, 6, 12 V 3, 4, 7
- 3, 10, 12 V 2, 4, 6
- 3, 6, 7 V 4, 10, 12
- 2, 7, 12 V 4, 6, 10
- 2, 3, 10 V 4, 7, 12
- 2, 7, 10 V 3, 6, 12

C

- 3, 4, 5 V 7, 8, 11
- 3, 7, 13 V 4, 5, 8
- 4, 11, 13 V 3, 5, 7
- 4, 7, 8 V 5, 11, 13
- 3, 8, 13 V 5, 7, 11
- 3, 4, 11 V 5, 8, 13
- 3, 8, 11 V 4, 7, 13

D

- 4, 5, 6 V 8, 9, 12
- 4, 8, 14 V 5, 6, 9
- 5, 12, 14 V 4, 6, 8
- 5, 8, 9 V 6, 12, 14
- 4, 9, 14 V 6, 8, 12
- 4, 5, 12 V 6, 9, 14
- 4, 9, 12 V 5, 8, 14

E

- 5, 6, 7 V 9, 10, 13
- 5, 9, 15 V 6, 7, 10
- 6, 13, 15 V 5, 7, 9
- 6, 9, 10 V 7, 13, 15
- 5, 10, 15 V 7, 9, 13
- 5, 6, 13 V 7, 10, 15
- 5, 10, 13 V 6, 9, 15

F

- 6, 7, 8 V 10, 11, 14
- 6, 10, 1 V 7, 8, 11
- 7, 14, 1 V 6, 8, 10
- 7, 10, 11 V 8, 14, 1
- 6, 11, 1 V 8, 10, 14
- 6, 7, 14 V 8, 11, 1
- 6, 11, 14 V 7, 10, 1

G

- 7, 8, 9 V 11, 12, 15
- 7, 11, 2 V 8, 9, 12
- 8, 15, 2 V 7, 9, 11
- 8, 11, 12 V 9, 15, 2
- 7, 12, 2 V 9, 11, 15
- 7, 8, 15 V 9, 12, 2
- 7, 12, 15 V 8, 11, 2

H

- 8, 9, 10 V 12, 13, 1
- 8, 12, 3 V 9, 10, 13
- 9, 1, 3 V 8, 10, 12
- 9, 12, 13 V 10, 1, 3
- 8, 13, 3 V 10, 12, 1
- 8, 9, 1 V 10, 13, 3
- 8, 13, 1 V 9, 12, 3

I

- 9, 10, 11 V 13, 14, 2
- 9, 13, 4 V 10, 11, 14
- 10, 2, 4 V 9, 11, 13
- 10, 13, 14 V 11, 2, 4
- 9, 13, 4 V 11, 13, 2
- 9, 10, 2 V 11, 14, 4
- 9, 13, 14 V 10, 13, 4

J

- 10, 11, 12 V 14, 15, 3
- 10, 14, 5 V 11, 12, 15
- 11, 3, 5 V 10, 12, 15
- 11, 14, 15 V 12, 3, 5
- 10, 15, 5 V 12, 14, 3
- 10, 11, 3, V 12, 15, 5
- 10, 15, 3 V 11, 14, 5

K

- 11, 12, 13 V 15, 1, 4
- 11, 15, 6 V 12, 13, 1
- 12, 4, 6 V 11, 13, 15
- 12, 15, 1 V 13, 4, 6
- 11, 1, 4 V 13, 15, 6
- 11, 12, 4 V 13, 1, 6
- 11, 1, 4 V 12, 15, 6

L

- 12, 13, 14 V 1, 2, 5
- 12, 1, 7 V 13, 14, 2
- 13, 5, 7 V 12, 14, 1
- 13, 1, 2 V 14, 5, 7
- 12, 2, 7 V 14, 1, 5
- 12, 13, 5 V 14, 2, 7
- 12, 2, 5 V 13, 1, 7

M

- 13, 14, 15 V 2, 3, 6
- 13, 2, 8 V 14, 15, 3
- 14, 6, 8 V 13, 15, 2
- 14, 2, 3 V 15, 6, 8
- 13, 3, 8 V 15, 2, 6
- 13, 14, 6 V 15, 3, 8
- 13, 3, 6 V 14, 2, 8

N

- 14, 15, 1 V 3, 4, 7
- 14, 3, 9 V 15, 1, 4
- 15, 7, 9 V 14, 1, 13
- 15, 3, 4 V 1, 7, 9
- 14, 4, 9 V 1, 3, 7
- 14, 15, 7 V 1, 4, 9
- 14, 4, 7 V 15, 3, 9

O

- 15, 1, 2 V 4, 5, 8
- 15, 4, 10 V 1, 2, 5
- 1, 8, 10 V 15, 2, 4
- 1, 4, 5 V 2, 8, 10
- 15, 5, 10 V 2, 4, 8
- 15, 1, 8 V 2, 5, 10
- 15, 5, 8 V 1, 4, 10

Triples: 19 Players 4 Reps / 1 Court

A

1. 1, 3, 5 V 6, 7, 8
2. 6, 7, 8 V 11, 14, 15
3. 11, 14, 15 V 1, 3, 5
4. 1, 6, 11 V 3, 7, 15
5. 1, 7, 14 V 3, 8, 11
6. 1, 8, 15 V 3, 6, 14
7. 3, 6, 14 V 5, 7, 11
8. 3, 7, 15 V 5, 18, 11
9. 3, 8, 11 V 5, 6, 15
10. 5, 6, 15 V 1, 7, 14
11. 5, 7, 11 V 1, 8, 15
12. 5, 8, 14 V 1, 6, 11

B

1. 2, 4, 6 V 7, 8, 9
2. 7, 8, 9 V 12, 15, 16
3. 12, 15, 16 V 2, 4, 6
4. 2, 7, 12 V 4, 8, 16
5. 2, 8, 15 V 4, 9, 12
6. 2, 9, 16 V 4, 7, 15
7. 4, 7, 15 V 6, 8, 12
8. 4, 8, 16 V 6, 9, 15
9. 4, 9, 12 V 6, 7, 16
10. 6, 7, 16 V 2, 8, 15
11. 6, 8, 12 V 2, 9, 16
12. 6, 9, 15 V 2, 7, 12

C

1. 3, 5, 7 V 8, 9, 10
2. 8, 9, 10 V 13, 16, 17
3. 13, 16, 17 V 3, 5, 7
4. 3, 8, 13 V 5, 9, 17
5. 3, 9, 16 V 5, 10, 13
6. 3, 10, 17 V 5, 8, 16
7. 5, 8, 16 V 7, 9, 13
8. 8, 3, 17 V 7, 10, 16
9. 5, 10, 13 V 7, 8, 17
10. 7, 8, 17 V 3, 9, 16
11. 7, 9, 13 V 3, 10, 17
12. 7, 10, 16 V 3, 8, 13

D

1. 4, 6, 8 V 9, 10, 11
2. 9, 10, 11 V 14, 17, 18
3. 4, 17, 18 V 4, 6, 8
4. 4, 9, 14 V 6, 10, 18
5. 4, 10, 17 V 6, 11, 14
6. 4, 11, 18 V 6, 9, 17
7. 6, 9, 17 V 8, 10, 14
8. 6, 10, 18 V 8, 11, 17
9. 6, 11, 14 V 8, 9, 18
10. 8, 9, 18 V 4, 10, 17
11. 8, 10, 14 V 4, 11, 18
12. 8, 11, 17 V 4, 9, 14

E

1. 5, 7, 9 V 10, 11, 12
2. 10, 11, 12 V 15, 18, 19
3. 15, 18, 19 V 5, 7, 9
4. 5, 10, 15 V 7, 11, 19
5. 5, 11, 18 V 7, 12, 15
6. 5, 12, 19 V 7, 10, 18
7. 7, 10, 18 V 9, 11, 15
8. 7, 11, 19 V 9, 12, 18
9. 7, 12, 15 V 9, 10, 19
10. 9, 10, 19 V 5, 11, 18
11. 9, 11, 15 V 5, 12, 19
12. 9, 12, 18 V 5, 10, 15

F

1. 6, 8, 10 V 11, 12, 13
2. 11, 12, 13 V 16, 19, 1
3. 16, 19, 1 V 6, 8, 10
4. 6, 11, 16 V 8, 12, 1
5. 5, 12, 19 V 8, 13, 16
6. 6, 13, 1 V 8, 11, 19
7. 8, 11, 19 V 10, 12, 16
8. 8, 12, 1 V 10, 13, 19
9. 8, 13, 16 V 10, 11, 1
10. 10, 11, 1 V 6, 12, 19
11. 10, 12, 16 V 6, 13, 1
12. 10, 13, 19 V 6, 11, 16

G

1. 7, 9, 11 V 12, 13, 14
2. 12, 13, 14 V 17, 1, 2
3. 17, 1, 2 V 7, 9, 11
4. 7, 12, 17 V 9, 13, 2
5. 7, 13, 1 V 9, 14, 17
6. 17, 14, 2 V 9, 12, 1
7. 9, 12, 1 V 11, 13, 17
8. 9, 13, 2 V 11, 14, 1
9. 9, 14, 17 V 11, 12, 2
10. 11, 12, 2 V 7, 13, 1
11. 11, 13, 17 V 7, 14, 2
12. 11, 14, 1 V 7, 12, 17

H

1. 8, 10, 12 V 13, 14, 15
2. 13, 14, 15 V 18, 2, 3
3. 18, 2, 4 V 8, 10, 12
4. 8, 13, 18 V 10, 14, 3
5. 8, 14, 2 V 10, 13, 18
6. 8, 15, 3 V 10, 13, 2
7. 10, 13, 2 V 12, 14, 18
8. 10, 14, 3 V 12, 15, 2
9. 10, 15, 18 V 12, 13, 2
10. 12, 13, 3 V 8, 14, 2
11. 12, 14, 18 V 8, 15, 3
12. 12, 15, 2 V 8, 13, 18

I

1. 9, 11, 13 V 14, 15, 16
2. 14, 15, 16 V 19, 3, 4
3. 19, 3, 4 V 9, 11, 13
4. 9, 14, 16 V 11, 15, 4
5. 9, 15, 3 V 11, 16, 19
6. 9, 16, 4 V 11, 14, 3
7. 11, 14, 3 V 13, 15, 19
8. 11, 15, 4 V 13, 16, 3
9. 11, 16, 19 V 13, 14, 4
10. 13, 14, 4 V 9, 15, 3
11. 13, 15, 19 V 9, 16, 4
12. 13, 16, 3 V 9, 14, 19

J

1. 10, 12, 14 V 15, 16, 17
2. 15, 16, 17 V 1, 4, 5
3. 1, 4, 5 V 10, 12, 14
4. 10, 15, 1 V 12, 16, 5
5. 10, 16, 4 V 12, 17, 1
6. 10, 17, 5 V 12, 15, 4
7. 12, 15, 4 V 14, 16, 1
8. 12, 16, 5 V 14, 17, 4
9. 12, 17, 4 V 14, 15, 5
10. 14, 15, 5 V 10, 16, 4
11. 14, 16, 1 V 10, 17, 5
12. 14, 17, 4 V 10, 15, 1

K

1. 11, 13, 15 V 16, 17, 18
2. 16, 17, 18 V 2, 5, 8
3. 2, 5, 8 V 11, 13, 15
4. 11, 16, 18 V 13, 17, 6
5. 11, 17, 5 V 13, 18, 2
6. 11, 18, 6 V 13, 16, 5
7. 13, 16, 5 V 15, 17, 2
8. 13, 17, 6 V 15, 18, 5
9. 13, 18, 2 V 15, 16, 6
10. 15, 16, 6 V 11, 17, 5
11. 15, 17, 2 V 11, 18, 6
12. 15, 18, 5 V 11, 16, 2

L

1. 12, 14, 16 V 17, 18, 19
2. 17, 18, 19 V 3, 6, 7
3. 3, 6, 7 V 12, 14, 16
4. 12, 17, 3 V 14, 18, 19
5. 12, 18, 5 V 14, 19, 3
6. 12, 19, 7 V 14, 17, 6
7. 14, 17, 6 V 16, 18, 3
8. 14, 18, 7 V 16, 19, 6
9. 14, 19, 3 V 16, 17, 7
10. 16, 17, 7 V 12, 18, 6
11. 16, 18, 3 V 12, 19, 7
12. 16, 19, 6 V 12, 17, 3

M

1. 13, 15, 17 V 18, 19, 1
2. 18, 19, 1 V 4, 7, 8
3. 4, 7, 8 V 13, 15, 17
4. 13, 18, 14 V 15, 19, 8
5. 13, 19, 7 V 15, 1, 4
6. 13, 1, 8 V 15, 18, 7
7. 15, 18, 7 V 17, 19, 4
8. 15, 19, 18 V 17, 1, 7
9. 15, 1, 4 V 17, 18, 8
10. 17, 18, 8 V 13, 19, 7
11. 17, 19, 4 V 13, 19, 8
12. 17, 1, 7 V 13, 18, 1

N

1. 14, 16, 18 V 19, 1, 2
2. 19, 1, 2 V 5, 8, 9
3. 5, 8, 9 V 14, 16, 18
4. 14, 19, 5 V 16, 1, 9
5. 14, 1, 8 V 16, 2, 5
6. 14, 2, 9 V 16, 19, 8
7. 16, 19, 8 V 18, 1, 5
8. 16, 1, 9 V 18, 2, 8
9. 16, 2, 5 V 18, 19, 9
10. 18, 19, 9 V 14, 1, 8
11. 18, 1, 5 V 14, 2, 9
12. 12, 18, 8 V 14, 19, 5

O

1. 15, 17, 19 V 1, 2, 3
2. 1, 2, 3 V 6, 9, 10
3. 6, 9, 10 V 15, 17, 19
4. 15, 1, 6 V 17, 2, 10
5. 15, 2, 9 V 15, 3, 6
6. 15, 3, 10 V 17, 1, 9
7. 17, 1, 9 V 19, 2, 6
8. 17, 2, 10 V 19, 3, 9
9. 15, 3, 6 V 19, 1, 10
10. 19, 1, 10 V 15, 2, 9
11. 19, 2, 6 V 15, 3, 10
12. 19, 3, 9 V 15, 1, 6

P

1. 16, 18, 1 V 2, 3, 4
2. 2, 3, 4 V 7, 10, 11
3. 7, 10, 11 V 6, 18, 1
4. 16, 2, 7 V 18, 3, 11
5. 16, 3, 10 V 18, 4, 7
6. 16, 4, 11 V 18, 2, 10
7. 18, 2, 10 V 1, 3, 7
8. 18, 3, 11 V 1, 4, 10
9. 18, 4, 7 V 1, 2, 11
10. 1, 2, 11 V 16, 3, 10
11. 1, 3, 7 V 16, 4, 11
12. 1, 4, 10 V 16, 2, 7

Q

1. 17, 19, 2 V 3, 4, 5
2. 3, 4, 5 V 8, 11, 12
3. 8, 11, 12 V 17, 19, 2
4. 17, 3, 8 V 19, 4, 12
5. 17, 4, 11 V 19, 5, 8
6. 17, 5, 2 V 19, 3, 11
7. 9, 3, 11 V 2, 4, 8
8. 19, 4, 12 V 2, 5, 11
9. 19, 5, 8 V 2, 3, 12
10. 2, 3, 12 V 17, 4, 11
11. 2, 4, 8, V 17, 5, 2
12. 2, 5, 11 V 17, 3, 8

R

1. 18, 1, 3 V 4, 5, 6
2. 4, 5, 6 V 9, 12, 13
3. 9, 12, 13 V 18, 1, 3
4. 18, 4, 9 V 1, 5, 13
5. 18, 5, 12 V 1, 6, 9
6. 18, 6, 13 V 1, 4, 12
7. 1, 4, 12 V 3, 5, 9
8. 1, 5, 13 V 3, 6, 12
9. 1, 6, 9 V 3, 4, 13
10. 3, 4, 13 V 18, 5, 12
11. 3, 5, 9 V 18, 6, 13
12. 3, 16, 12 V 18, 4, 9

S

1. 19, 2, 4 V 5, 6, 7
2. 5, 6, 7 V 10, 13, 14
3. 10, 13, 14 V 19, 2, 4
4. 19, 5, 10 V 2, 6, 14
5. 19, 6, 13 V 2, 7, 10
6. 19, 7, 14 V 2, 5, 13
7. 2, 5, 13 V 4, 6, 10
8. 2, 6, 14 V 4, 7, 13
9. 2, 7, 10 V 4, 5, 14
10. 4, 5, 14 V 19, 6, 13
11. 4, 6, 10 V 19, 7, 14
12. 4, 7, 13 V 19, 5, 10

TOURNAMENT SCORESHEETS

RANKING TOURNAMENT: DOUBLES FOR 12 (on two courts)

NO.	NAMES	SCORES (+ or -)																WINS	LOSSES	TOTAL	RANK
		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q			
1																					
2																					
3																					
4																					
5																					
6																					
7																					
8																					
9																					
10																					
11																					
12																					

SCHEDULE:

GAME

COURT I

COURT II

A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
Q

1, 2 V 6, 7
5, 9 V 8, 11
1, 3 V 7, 8
6, 10 V 9, 12
1, 4 V 8, 9
7, 11 V 2, 10
1, 5 V 9, 10
1, 6 V 10, 11
1, 7 V 11, 12
1, 8 V 2, 12
4, 11 V 3, 6
1, 9 V 2, 3
5, 12 V 4, 7
1, 10 V 3, 4
1, 11 V 4, 5
1, 12 V 5, 6
2, 6 V 5, 8

3, 12 V 4, 10
4, 6 V 2, 7
2, 4 V 5, 11
5, 7 V 3, 8
3, 5 V 6, 12
6, 8 V 4, 9
8, 12 V 3, 11
2, 9 V 4, 12
3, 10 V 2, 5
7, 9 V 5, 10
10, 12 V 2, 8
8, 10 V 6, 11
2, 11 V 3, 9
9, 11 V 7, 12
3, 7 V 6, 9
4, 8 V 7, 10

RANKING TOURNAMENT: DOUBLES FOR 12 (on three courts)

NO.	NAMES	SCORES (+ or -)											WINS	LOSSES	TOTAL	RANK
		A	B	C	D	E	F	G	H	I	J	K				
1																
2																
3																
4																
5																
6																
7																
8																
9																
10																
11																
12																

SCHEDULE:

GAME

COURT I

COURT II

COURT III

A	1, 2 V 6, 7	3, 12 V 4, 10	5, 9 V 8, 11
B	1, 3 V 7, 8	2, 4 V 5, 11	6, 10 V 9, 12
C	1, 4 V 8, 9	3, 5 V 6, 12	7, 11 V 2, 10
D	1, 5 V 9, 10	4, 6 V 2, 7	8, 12 V 3, 11
E	1, 6 V 10, 11	5, 7 V 3, 8	2, 9 V 4, 12
F	1, 7 V 11, 12	6, 8 V 4, 9	3, 10 V 2, 5
G	1, 8 V 2, 12	7, 9 V 5, 10	4, 11 V 3, 6
H	1, 9 V 2, 3	8, 10 V 6, 11	5, 12 V 4, 7
I	1, 10 V 3, 4	9, 11 V 7, 12	2, 6 V 5, 8
J	1, 11 V 4, 5	10, 12 V 2, 8	3, 7 V 6, 9
K	1, 12 V 5, 6	2, 11 V 3, 9	4, 8 V 7, 10

RANKING TOURNAMENT: DOUBLES FOR 16 (on four courts)

NO.	NAMES	SCORES (+ OR -)																WINS	LOSSES	TOTAL	RANK
		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O					
1																					
2																					
3																					
4																					
5																					
6																					
7																					
8																					
9																					
10																					
11																					
12																					
13																					
14																					
15																					
16																					

SCHEDULE:

GAME	COURT I	COURT II	COURT III	COURT IV
A	1, 2 V 7 - 12	3, 4 V 6, 10	5, 13 V 14, 16	8, 9 V 11, 15
B	1, 7 V 2, 12	3, 6 V 4, 10	5, 14 V 13, 16	8, 11 V 9, 15
C	1, 12 V 2, 7	3, 10 V 4, 6	5, 16 V 13, 14	8, 15 V 9, 11
D	1, 3 V 8, 13	4, 5 V 7, 11	2, 6 V 14, 15	9, 10 V 12, 16
E	1, 8 V 3, 13	4, 7 V 5, 11	2, 14 V 6, 15	9, 12 V 10, 16
F	1, 13 V 3, 8	4, 11 V 5, 7	2, 15 V 6, 14	9, 16 V 10, 12
G	1, 4 V 9, 14	5, 6 V 8, 12	3, 7 V 15, 16	2, 10 V 11, 13
H	1, 9 V 4, 14	5, 8 V 6, 12	3, 15 V 7, 16	2, 11 V 10, 13
I	1, 14 V 4, 9	5, 12 V 6, 8	3, 16 V 7, 15	2, 13 V 10, 11
J	1, 5 V 10, 15	6, 7 V 9, 13	2, 4 V 8, 16	3, 11 V 12, 14
K	1, 10 V 5, 15	6, 9 V 7, 13	2, 8 V 4, 16	3, 12 V 11, 14
L	1, 15 V 5, 10	6, 13 V 7, 9	2, 16 V 4, 8	3, 14 V 11, 12
M	1, 6 V 11, 16	7, 8 V 10, 14	2, 3 V 5, 9	4, 12 V 13, 15
N	1, 11 V 6, 16	7, 10 V 8, 14	2, 5 V 3, 9	4, 13 V 12, 15
O	1, 16 V 6, 11	7, 14 V 8, 10	2, 9 V 3, 5	4, 15 V 12, 13

[illegible]

[illegible][illegible][illegible]

RANKING TOURNAMENT: TRIPLES FOR NINES																		
NO.	NAMES	SCORES (+ or -)												WINS	LOSS	TOTAL	RANK	SCHEDULE
		A	B	C	D	E	F	G	H	I	J	K	L					
1																	A	1-4-7 V 2-5-9
2																	B	1-5-8 V 2-6-7
3																	C	1-6-9 V 2-4-8
4																	D	2-4-8 V 3-5-7
5																	E	2-5-9 V 3-6-8
6																	F	2-6-7 V 3-4-9
7																	G	3-4-9 V 1-5-8
8																	H	3-5-7 V 1-6-9
9																	I	3-6-8 V 1-4-7
																	J	1-2-3 V 4-5-6
																	K	4-5-6 V 7-8-9
																	L	7-8-9 V 1-2-3

APPENDIX

APPENDIX

The following is original information provided by James H. Bjerring. Used by permission.

Introduction

If a doubles tournament for 'n' players could be designed such that:

- (i) each player has each other player for a partner exactly once
- (ii) each player has each other player for an opponent exactly twice

the results for the 'n' players would be comparable and so a linear ranking of the players could be obtained.

Unfortunately, the design of such tournaments is usually non-trivial and often impossible, depending on the value of 'n'. For example, since a total of 'n' (n-1) / 4 games must be played, the remainder must be 0 or 1 when 'n' is divided by 4. This restriction eliminates half the values of 'n' immediately.

It is possible to design successful tournaments using three different methods which are listed below in ascending order of difficulty:

- (i) Orthogonal Latin Squares
- (ii) Balanced Incomplete Blocks
- (iii) Rouse Ball's Method (see reference 1)

Only the first (and easiest) method is described here, and then only for the special case of sixteen players. However, complete tournament designs for 4, 5, 8, 12, 16, and 25 players are given in the following pages.

Special Doubles Rules

1. Your receivers must be rotated.
2. You can win by one point.
3. The ball may cross the net at any point. Landing in the court is what counts. (the above rules only apply to a Jim Bjerring Tournament).

Scoring

With each player is associated two numbers. The first is the number of games he won. The second is the total number of points he scored minus the total number of points scored against him. A lexicographical (componentwise) sort of the sixteen pairs of numbers yields a linear ranking of the 16 players. Winners are decided by games won. The over-all point difference is used in the case of ties.

How to Use the Doubles Tournament for Ranking Your Players

This type of tournament has been used at the National Women's volleyball training camp. The results indicated that this kind of tournament has a very high validity for ranking players.

1. Use a court 4 meters wide and 18 meters long.
2. For 16 teams, you can run a complete session in 3 hours with games to 7 points. If you wish, it is better to play to 11 or 15 points and extend over 2 or 3 practices.

Features of Special Interest

The narrow court is very effective for observing all the skills.
The only skill missed is the long volley.

Coaches will note that during the tournament is a good time to evaluate individual skills. You get a chance to see them use all their skills.

LINEARDOUBLES AND TRIPLES
RANKING TOURNAMENTS**TOURNEY SHEET**

30

Group: _____ Activity: _____

Day/Date/Round:

1.						
2.						
3.						
4.						
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50.						

SAMPLE SCORECARD (Eight Players)

Player: Game	1	2	3	4	5	6	7	8	Totals
1									
2									
3									
4									
5									
6									
7									
No. of Wins									
Total Points									

In the (i, j)th cell put the number of points scored by player j in game i minus the number of points scored against player j in game i. Note that a '+' denotes a victory and a '-' denotes a loss.

Checks:

1. The sum of each of the first 7 rows should be zero.
2. The sum of the "number of wins" row should be 28.
3. The sum of the "total points" row should be zero.

SAMPLE SCORESHEET AND DISCUSSION

NO.	NAMES	SCORES (+ or -)					RANKING TOURNAMENT: DOUBLES FOR FIVE					SCHEDULE
		A	B	C	D	E	WINS	LOSS	TOTAL	RANK		
1	BILL	- 7	+ 12	- 2	-4	OUT	1	3	- 1	4	A 1-2 V 3-4	
2	JANE	- 7	- 12	+ 2	OUT	+7	2	2	- 10	3	B 1-3 V 2-5	
3	JO	+ 7	+ 12	OUT	+ 4	+ 7	4	0	+ 30	1	C 1-5 V 2-4	
4	SAM	+ 7	OUT	+ 2	- 4	- 7	2	2	- 2	2	D 1-4 V 3-5	
5	CHUCK	OUT	- 12	- 2	+ 4	- 7	1	3	- 17	5	E 2-3 V 4-5	

In the sample above there are a number of items of particular interest.

In match-up "A", the margin of difference between the winners; Jo and Sam, and the losers; Bill and Jane, was seven points and of course the winners wrote in a plus seven and the losers registered a negative. Chuck was out and he officiated (as all "out" players are expected to do).

There is no need to go through every pairing, but notice that in match-up "B", as sometimes happens, there was a huge differential between winners and losers... twelve big points! Such a difference has a great impact on the final outcome of total points - obviously, but there is more to it than just that. Much of the time players seem to forget that their win - loss record is most important and they get caught up in their point total. Whenever one game has a big margin of difference the totals may seem **not** to reflect the ranking from Win - Loss. Look at the second, third and fourth ranked players: Sam, Jane and Bill respectively. If ranking was based on points total, Bill would rank second instead of fourth. The implications are clear. First, winning is what matters most when competing in a ranking tourney. And second, every pairing and every game is important because the differential will count either for or against each competitor.

Finally, any time that one player goes undefeated there will be two or more players who have identical win - lose records. So, if a group has an outstanding player in it, and he or she are likely to win every match-up; then everybody else better compete like maniacs for their own critical point differences!